BIOT 5206 · Spring 2026 · Term Project

Hippocampal aging across the lifespan

A microarray reanalysis of Kadish et al. (2009)

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GSE9990 · 49 rats · 5 age groups · ~15,900 probes · rat hippocampus CA1

The biological question

How does the hippocampus age — and when does the transition happen?

Cognitive decline begins in middle age, long before clinical dementia. But what changes inside the brain — and at which life stage — to drive this decline?

Kadish et al. (2009, J. Neurosci.) profiled rat hippocampal gene expression at 5 age points to identify when transcriptional programs of aging activate.

This project: reproduce their findings using modern statistical methodology — and push further with co-expression network analysis.

49

male F344 rats, single hippocampus sample each

5

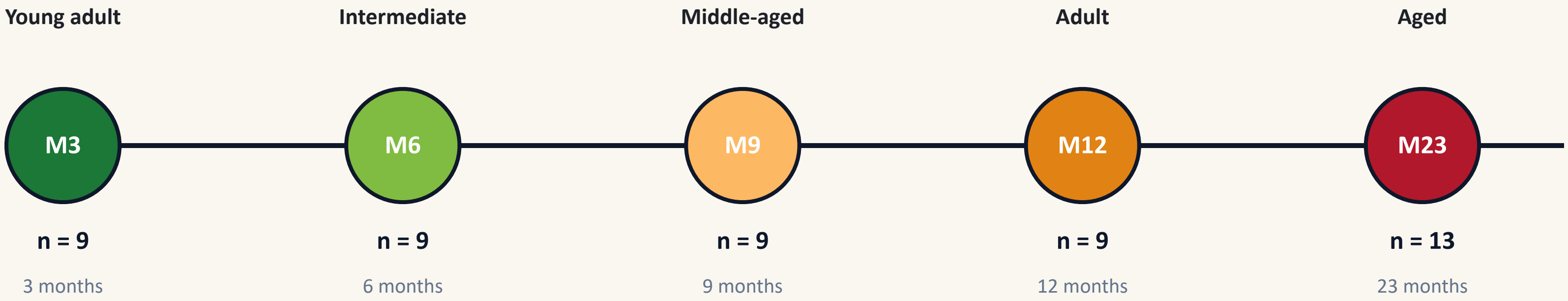
age groups: M3, M6, M9, M12, M23 months

Affymetrix RAE230A

~15,900 annotated probe sets
CA1 region, cross-sectional design

Experimental design

A cross-sectional view of the rat lifespan



Total samples	Tissue	Design	RNA per rat
49	**Hippocampus CA1**	**Cross-sectional**	**Single sample**

Analysis pipeline

Eight phases mirroring the Lecture 8 / IRF6 workflow, adapted for time-course data

0 Environment limma, rae230a.db, clusterProfiler	**1 Load + log2** QQ plot before & after transform	**2 Quality control** Boxplots, clustering, PCA, per-PC ANOVA	**3 F-test** limma with eBayes across all 5 ages
4 Heatmap Top 100 ARGs visualized by age	**5 Template matching** Assign onset × direction categories	**6 Enrichment** GO BP with array-specific background	**7 Anchor genes** 10 canonical aging markers

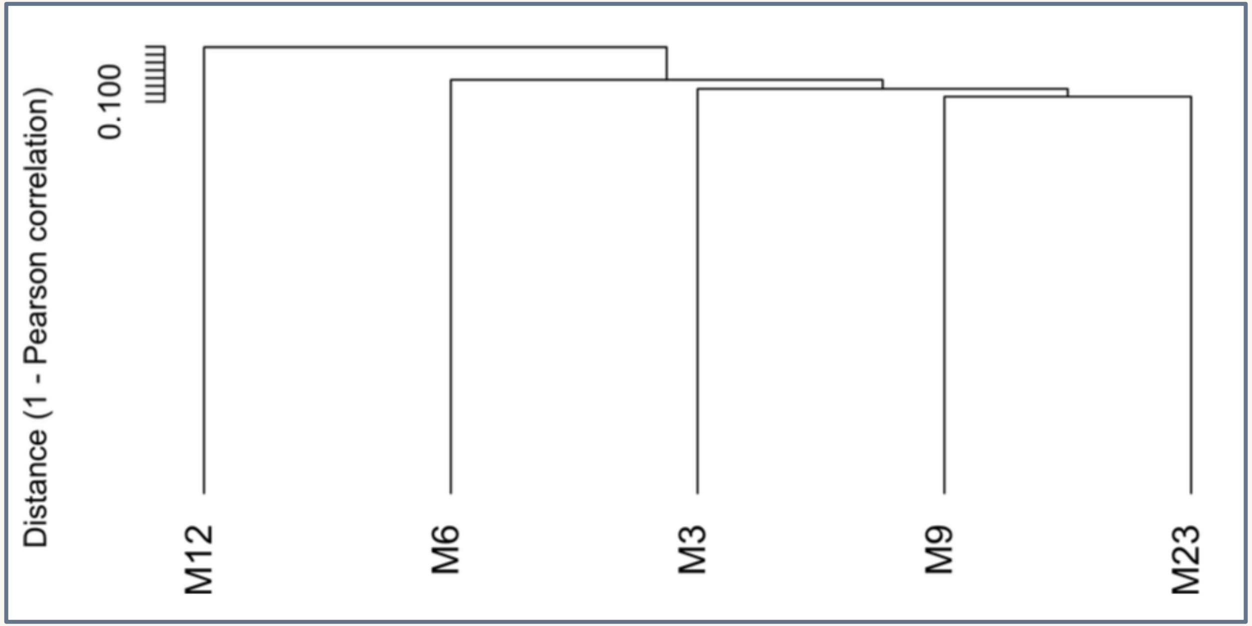
Bonus

Differential co-expression networks

Network rewiring: M3 vs M23 (total lifespan) and M12 vs M23 (late-life-specific)

Quality control

Hierarchical clustering – Top 2000 variable genes, Pearson distance

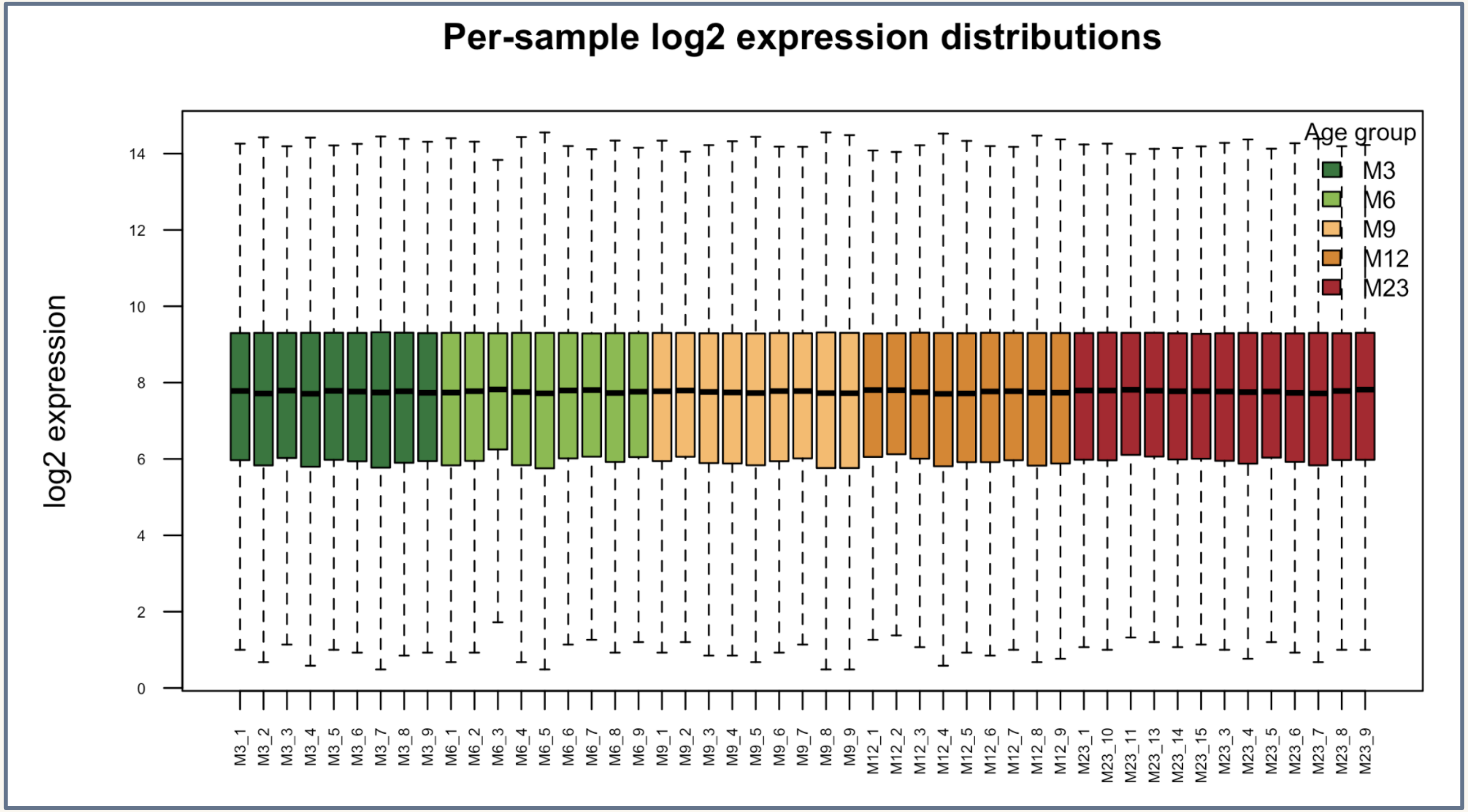


Conclusion

M9 clusters with M3/M6, while M12 clusters with M23 — the major transition occurs between M9 and M12. This is the midlife inflection point.

Quality control

Box-plot – Showing the distribution of samples



Differential expression: the F-test

limma with empirical Bayes shrinkage, BH-corrected FDR

358

aging-related genes

at $q < 0.25$ (Kadish FDR)

157 pass at $q < 0.05$

vs. original paper

923 ARGs reported by Kadish 2009

Why fewer ARGs?

- Limma is more conservative than plain ANOVA
- Public matrix is pre-filtered (~15,900 vs original ~22,000 probes)
- Annotation has been updated since 2009

Sanity check: canonical aging markers

Recovered (5 / 9):

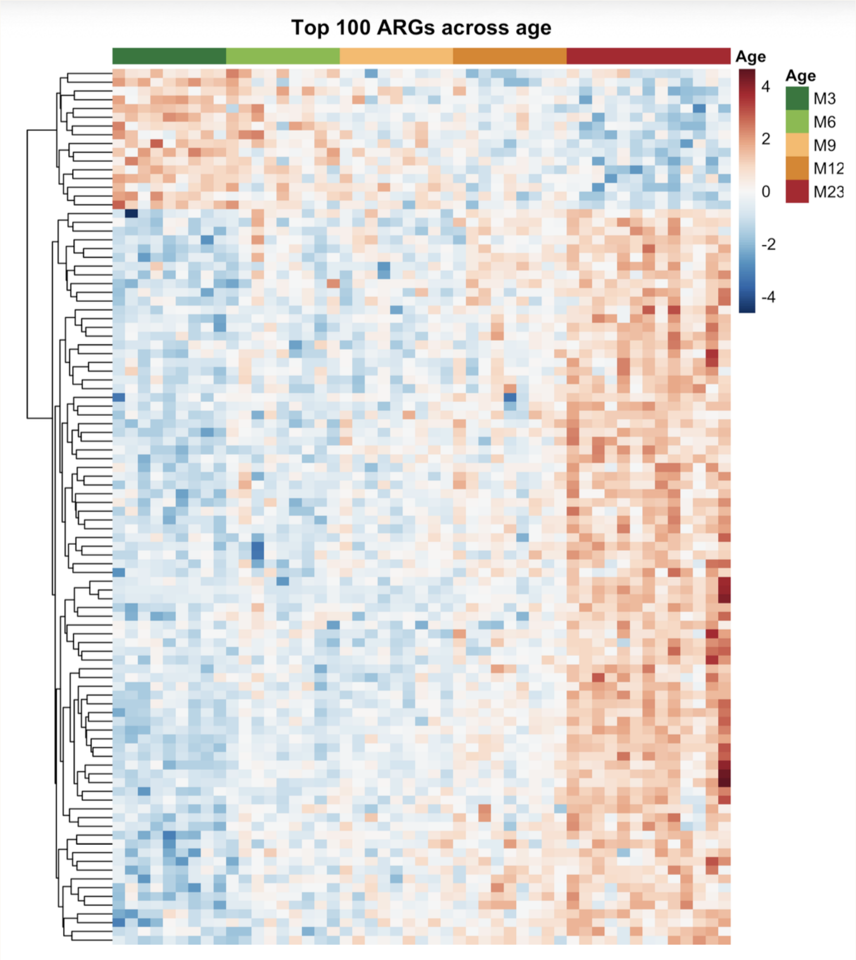
Gfap | *Ctsd* | *C1qb* | *S100a4* | *B2m*

Not in ARG list (4 / 9):

Apoe | *App* | *Mog* | *Mobp*

Apoe shows a clear age-dependent rise in the anchor gene panel — it just falls outside the FDR cutoff due to limma's stricter shrinkage. The biology is captured even when individual probes don't cross threshold.

Differential expression: Heatmap



Temporal patterns

Template matching assigns each ARG to an onset × direction category — when does each gene change?

309

of 358 ARGs assigned (86%)

Templates correlate each gene's mean profile against 10 idealized shapes; we keep matches with |r| ≥ 0.87.

Early

ONSET

Changes between M3 and M6

90 up-regulated

30 down-regulated

120

total genes

Intermediate

ONSET

Starts changing at M9

38 up-regulated

12 down-regulated

50

total genes

Midlife

ONSET

Activates at M12 — the inflection

59 up-regulated

12 down-regulated

71

total genes

Late

ONSET

Only changes by M23

43 up-regulated

25 down-regulated

68

total genes

Up-regulation dominates 3:1 — aging primarily activates immune and inflammatory programs, rather than silencing constitutive functions.

The biological cascade

GO enrichment reveals when each aging program activates — exactly as Kadish described

EARLY

M3 → M6

Immune activation begins

Top GO terms

- Antigen processing
- Immune response
- MHC class I presentation

INTERMEDIATE

M6 → M9

Bioenergetic transition

Top GO terms

- Reactive oxygen species
- Leukocyte activation
- Myeloid cell activation

MIDLIFE

M9 → M12

Astrocyte program

Top GO terms

- Astrocyte activation
- Astrocyte development
- Macrophage migration

LATE

M12 → M23

Expanded inflammation

Top GO terms

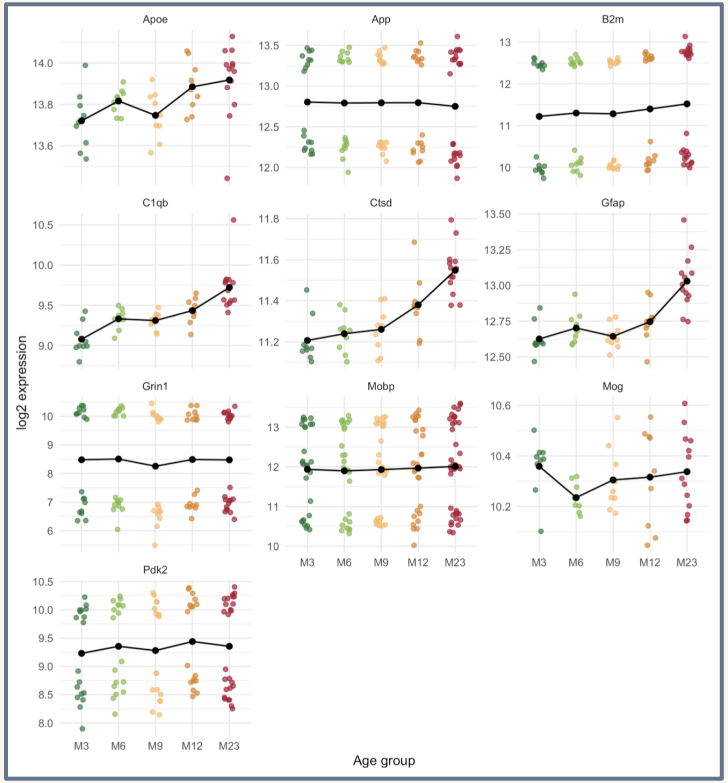
- T cell activation
- Cell adhesion
- MHC class II presentation

Down-regulated late-onset:

Vesicle transport · Synaptic vesicle recycling · Microtubule-based movement — the synaptic decline of aging

Anchor gene trajectories

Individual gene plots concretely confirm the aging biology



What the panel shows

Astrocyte cluster rises in lockstep

Apoe, Gfap, Ctsd, C1qb all up-regulate together — steepest rise between M9 and M12, exactly the midlife inflection.

Myelin genes track late aging

Mog and Mobp show smaller but consistent late-life increases, suggesting myelin remodeling continues into old age.

Synaptic gene shows expected decline

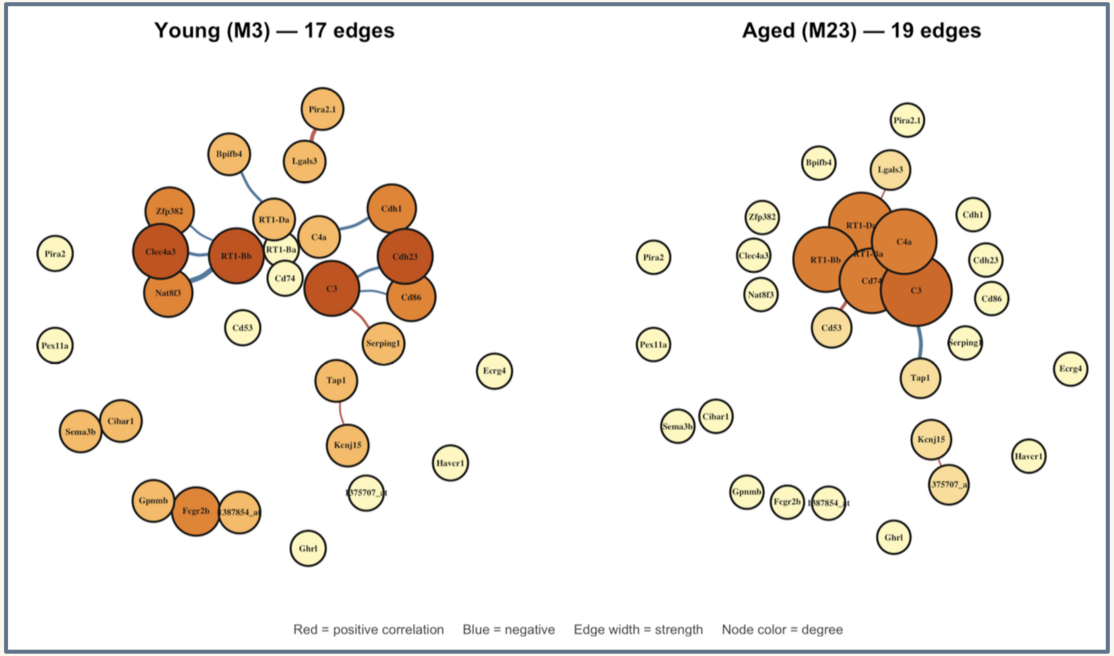
Grin1 (glutamate receptor) trends downward with high variance — classical synaptic dysfunction signature.

App is transcriptionally flat

Suggesting amyloid-precursor biology in aging is post-transcriptional, not mRNA-level.

Bonus: Aging rewires the co-expression network

Differential co-expression analysis of top 30 ARGs



The finding

0

edges preserved

between M3 and M23 networks

The transcriptional network is essentially completely rewired between young and aged animals.

Comparison with Kadish et al. (2009)

Three big agreements, three explained discrepancies

Where we agree

Temporal cascade

Same 4 onset categories (early / intermediate / midlife / late) emerge from independent statistical methodology.

Midlife inflection at M9 → M12

Confirmed by hierarchical clustering AND anchor gene trajectories — dual independent validation.

Canonical markers recovered

Gfap, Ctsd, C1qb, S100a4, B2m all appear in ARG list with expected direction of change.

Where we differ — and why

358 ARGs vs 923 reported

limma's eBayes shrinkage is stricter than plain ANOVA. Pre-filtered probe set further reduces input.

Apoe, App, Mog, Mobp not in ARG list

Yet they show clear age trajectories in the anchor panel — biology preserved, but stricter test fails them at FDR.

Different statistical method

limma F-test with empirical Bayes vs original ANOVA. Better variance estimation, harder threshold to cross.

Limitations

Honest accounting of what this analysis cannot conclude

DATA

Pre-filtered expression matrix

The public TSV contains ~15,900 probes vs the original ~22,000 on RAE230A. Probes filtered upstream may have carried real aging signal — we can't recover those genes.

STATISTICAL

Stricter test than the original

limma's eBayes shrinkage is more conservative than plain ANOVA. Genes with higher residual variance (Apoe, App, Mog, Mobp) fail FDR despite showing clear trajectories.

DESIGN

Cross-sectional, not longitudinal

Each rat contributes one sample at one age. We cannot distinguish age effects from cohort effects — only follow-up longitudinal studies could resolve this.

NETWORK

Small per-group sample size

n=9 (M3) and n=13 (M23) make correlation estimates noisy. Bootstrap addresses this partially, but formal differential co-expression methods (e.g., DiffCorr) would be more rigorous.

These limitations don't invalidate the findings — they define the scope of what we can and can't conclude.

CLOSING THOUGHTS

Thank you.

The hippocampus ages in coordinated phases — and we can read those phases in the transcriptome.

Questions?

References

Kadish et al. (2009) J. Neurosci. 29(6):1805-16 · GEO GSE9990 · Affymetrix RAE230A

Tools: limma, clusterProfiler, igraph, pheatmap, ggplot2 in R 4.3+